Steering a 3B Code Model Selects One Known Algorithm Rather Than Diversifying

Executive summary of the run

16 October 2026

1. Headline

Away-from-mode steering selects one predicted algorithm family rather than flattening the distribution. In a pre-registered test on Qwen2.5-3B-Instruct, the test's rules, strengths and predicted gainers were frozen by hash before the main runs. A difference-in-means direction pointing away from the model's default algorithm met the frozen SELECT rule on both tasks, on the main draw and on a fresh-seed re-draw. On sorting in the re-draw (seeds 1000 to 1149, strength s2, $\alpha = 0.473$, n = 150), bubble sort fell from 62% of programs to 0% and quicksort, the predicted gainer, rose to 68%. Entropy over algorithm families was 1.022 bits against a mean of 1.749 bits over five matched-norm random directions. The difference is -0.727 bits, 95% CI $[-0.954, -0.451]$ (Miller-Madow entropy, hierarchical bootstrap over draws and samples, 10,000 resamples). The pass rate was 0.913 against 0.984 for the random-draw mean. *Source:* artifact gen_art_experiment_6, round 2 (report Table 73). We checked these values against the artifact's `results/selector_tables.json` and they match.

Evidence grade. The frozen rule's verdict is SELECT on both tasks and both draws. The round-2 record grades this strand *lead*, not confirmed: review score 5, non-blocking. **Conflict:** the paper states the result as a passed pre-registered test, while the round record keeps it at "lead" because of three caveats. The direction built for the other task also moves path search to A* (40 of 150). Individual random draws also move sorting a lot (random mode share 0.19 to 0.69). And the path-search operating point missed its frozen pass-rate bar (pass 0.17, against 0.31 at baseline).

Scope. One 3B model, the HF engine, one bare prompt wording per task, two tasks. The SELECT rule accepted the top gainer of either of two frozen rankings; the projection ranking was right on both tasks and the nearest-neighbour ranking was wrong on both. Selection reaches only ideas the model already writes.

2. Goal

The run asked how an instruction-tuned open-weight model chooses the solution idea for an open coding task (for example, which path-search or sorting algorithm it writes), and whether that choice can be pushed toward ideas the model does not usually produce. The knobs were temperature and top-k, random seed strings, prompt instructions (bans, mentions, novelty requests) and internal edits (activation steering, concept erasure). Every program was labelled by an audited classifier and checked by execution, so that a change of idea could be told apart from a change of wording, and a correct new program from a broken one.

3. What was tried

Round 1. A pre-registered three-arm screen on Qwen2.5-3B-Instruct (temperature, bans, internal knobs), with survival rules frozen before data existed. Only the influence arm, gen_art_experiment_3, was analysed at the time: 19,250 generations comparing steering and erasure with temperature, novelty prompts, a top-3 ban and generate-then-select, on two path tasks whose prompts carried a fixed output contract. No internal knob survived. The reviewer blocked the round (score 4) for two reasons: the other two arms had left thousands of unanalysed generations on disk, and the output contract had narrowed each task to a two-idea menu.

Round 2. All core tests moved to the user's bare prompt. gen_art_experiment_4 labelled and executed 13,968 programs with one audited classifier, wrote up the round-1 ban arm, and tested a frozen temperature law and a frozen seed-shock predictor

on fresh draws. gen_art_experiment_5 treated bans as displacements of the prompt-final residual stream: it froze seven substitution predictors and scored them on 5,930 never-generated confirmation programs (including a new k-th-smallest task), then ran transplant and attention-knockout tests. gen_art_experiment_6 re-analysed the round-1 random-control claim, ran the pre-registered select-or-flatten test, and repeated the ban and temperature tests on gemma-2-2b-it. The reviewer scored the round 5, non-blocking.

4. Key results

Finding	Numbers	Source	Grade (round record)
Steering away from the default selects the predicted idea (headline)	Sorting re-draw: entropy -0.727 $[-0.954, -0.451]$ vs random mean; quicksort 65%/68%. Path search: A* 57%/63%, pass 0.17 vs 0.337 (random mean)	gen_art_exp._6, round 2	frozen rule met; strand "lead"
The bare-prompt default is the mode of a small menu	Path search: BFS 179, DFS 44, A* 39, Dijkstra 22 of 300; weighted task: Dijkstra 300 of 300	gen_art_exp._4, round 2	descriptive
Temperature adds ideas slowly and breaks code	Path search, T 0.3 to 1.0: entropy 1.698 to 1.962 bits, pass 0.49 to 0.19; frozen temperature law met on no task	gen_art_exp._4, round 2	law failed (pre-registered)
Seed strings narrow the ideas	Entropy change at T = 0.7: -0.337 $[-0.529, -0.132]$ (path), -0.677 $[-0.917, -0.467]$ (sorting)	gen_art_exp._4, round 2	"lead"; predictor failed
Bans lower ideas they do not name	8 of 46 cells at Holm $p < 0.01$, 0 of 45 placebo-vs-placebo; banning greedy best-first: A* 3% to 58%	gen_art_exp._5, round 2	passed (pre-registered); "lead"
Internal displacement barely beats proportional renormalisation	Mean JSD 0.188 vs 0.207 (IIA): 9.2% gain $[-0.6\%, 19.6\%]$, 20% required; sign accuracy on T1–T3 bans 0.795 vs 0.591 (0.83 vs 0.65 over all 52 sign cells)	gen_art_exp._5, round 2	failed (pre-registered)
The prompt-final state does not carry the ban	Transplant recovery -0.001 vs -0.006 (random); ban-sentence knockout restoration 0.719	gen_art_exp._5, round 2	transplant failed; knockout exploratory
Asking beats steering for off-menu ideas	Steering toward IDDFS, Bellman-Ford, counting sort: 0.00 to 0.01 target share; asking: 0.92 to 1.00; heap sort steered 0.91	gen_art_exp._6, round 2	"lead"
Correct off-menu programs are rare	≤ 0.08 at any temperature; sorting top-3 ban 44 of 100; path search ≤ 1 of 100 under any knob	gen_art_exp._4/_6, round 2	passed on sorting only
Second model: displacement predictor loses	gemma-2-2b-it, 14 held-out bans: mean JSD 0.121 vs 0.088 (IIA)	gen_art_exp._6, round 2	failed
Contract-prompt steering also selects	Difference-in-means $\alpha = 0.224$: DFS 145 of 150 at pass 0.680; no knob met its survival rule	gen_art_exp._3, round 1	"lead" (contract menu)

5. What failed

- **Flattening as a novelty knob** (round 1). Directions meant to flatten the idea utilities instead selected the minority idea. The held-out entropy ratio against temperature was 0.932 and 1.203, where 2 was required.
- **Menu erasure and the cross-task "unusualness dial"** (round 1). LEACE removed idea decodability, yet the outputs stayed BFS, DFS and Dijkstra. The dial was not monotone in strength (Spearman 0.486 and -0.257, exact p 0.356 and 0.658). No program the judge accepted as a novel correct algorithm appeared in any condition (0 of 70 checked).
- **The pre-registered temperature law** (round 2). It was met on no task and draw: path-search intercepts reject pure temperature scaling, and the frozen law lost to fixed utilities on sorting out of sample.
- **Seed strings as predictable shocks** (round 2). On sorting the predicted greedy flip rate was 0.394 and the observed rate 0.017; the internal readout failed its coverage gate.
- **Internal geometry as the predictor of substitution** (round 2). The gain over IIA was 9.2% where 20% was required, top-substitute hits were 20% where 70% was required, and the predictor was worse than IIA on the k-th-smallest task and on gemma.
- **The prompt-final state as the carrier of a ban** (round 2). Transplanting, subtracting or overwriting it moved nothing, on both models.
- **Two round-1 claims withdrawn.** The claim that random directions raise entropy as much as targeted ones fails a draw-averaged comparison (3 targeted settings beat the draw mean, 5 fall below it, 10 are undetermined). The claim that "bans are ignored" holds only under the contract prompt with list wording.

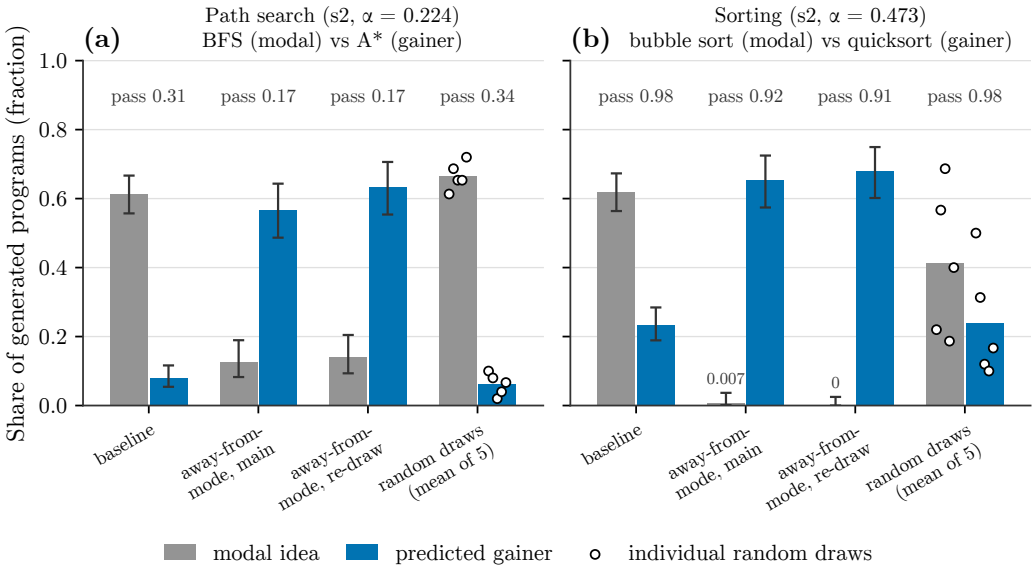


Figure 1: The headline test (gen_art_experiment_6, round 2). Share of programs implementing the modal idea (grey) and the frozen predicted gainer (blue) at strength s2, for the baseline, the away-from-mode direction on the main draw and the fresh-seed re-draw, and the mean of five matched-norm random directions (open circles: individual draws). Error bars are Wilson 95% intervals; the oracle pass rate is printed above each group.

6. Open questions

- Where the idea is committed. Forking-path resampling was planned in both rounds and run in neither.
- Whether the ban works as a prohibition or as name priming. Every regularity-violating ban names an in-domain algorithm, and the neutral-mention control was not run.
- How the ban sentence is read during decoding. The knockout rests on four bans at n = 60. Knocking out the placebo sentence also shifts output, and no random-span control was run.
- How specific the selector is. The off-target direction also raises A*, and some random draws select on sorting.
- Scope. One wording per condition and one Qwen size; gemma's path-search labels are label-limited (kappa 0.39); the user's two literal seed strings were never tested.

The full chronological record, with every table, is the 112-page internal report (`report.pdf`).